LOKNETE MA. HANMANTRAO PATIL CHARITABLE TRUST'S
ADARSH INSTITUTE OF TECHNOLOGY AND RESEARCH CENTRE, VITA
----------An Autonomous Institute----------
(NAAC & NBA Accredited Institute)
A/P: Khambale(Bha) Near Karve MIDC, Vita **Tal:** Khanapur **Dist:** Sangli.415311
Phone: 7755900777 Email: aitrc@agiv.edu.in **Web :** www.aitrcvita.edu.in
Hon.Adv.Sadashivrao H Patil Ex.MLA. Founder
Hon.Adv.Vaibhav S Patil President

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Academic Year: 2025-26	Mini Project-I Synopsis		Sem-V
	Department : Computer Science and Engineering		Date of Preparation: 10/09/2025
Roll No	3029,3030,3031,3032,3043	Class	T.Y.CSE
Domain Name	Artificial Intelligence and Machine Learning, Computer Vision		
Project Title	Smart Queue Monitoring System		
Student Name	1.Satyajeet.D.More 2.Onkar.K.Patil 3.Prachi.I.Sakate 4.Prashant.K.Sawant 5.Sadabashya.A.Mujawar		
Guide Name	Ms. P. R. Telshinge		

Introduction:

Queues are very common part of daily life in banks, hospitals, shopping malls, airports, government offices, and various service-oriented sectors. Managing these queues effectively is a crucial factor for ensuring customer satisfaction and operational efficiency. In many places, customers spend unnecessary time waiting without having clarity about the length of the queue or their expected waiting time. This often leads to frustration, dissatisfaction, and poor overall user experience.

Traditional queue management solutions such as manual monitoring, token-based systems, or simple CCTV surveillance do not provide **real-time actionable insights**. While these methods can record the number of people entering or leaving, they are not efficient in continuously tracking and updating queue length, especially during peak hours or in crowded environments. Moreover, they often require human intervention, which increases cost and reduces efficiency.

With the advancement of Artificial Intelligence and Computer Vision, it has become possible to automate the monitoring of queues with high accuracy. Deep learning models such as **YOLO (You Only Look Once)** can detect multiple people in real-time from camera feeds. Combined with frameworks like **OpenCV** for image processing and backend systems such as **Flask/FastAPI**, an intelligent system can be developed that not only detects people in a queue but also calculates important statistics like:

- Queue length
- Average waiting time per person
- Estimated waiting time for the next customer

Thus, this project focuses on building an **AI-powered Queue Detection and Management System** that integrates **Computer Vision, Deep Learning, and Web Technologies**.

It aims to bridge the gap between research-focused people detection systems and practical real-time queue monitoring applications that can enhance customer experience and help organizations optimize their operations.

Literature Review :

1. **Traditional Queue Management**: Token-based or manual systems are widely used but lacks real-time monitoring.
2. **Computer Vision in Queue Detection**: Studies have applied face/person detection algorithms but often suffer from accuracy issues in crowded conditions.
3. **Deep Learning Approaches**: Recent research shows YOLOv5/v8 and other deep learning models achieve high accuracy in real-time person detection and crowd counting.
4. **Existing Gaps**: Most existing work focuses on people counting, but fewer focus on integrating results with real-time dashboards for practical usage.

Relevance of the Work:

- Growing demand for smart infrastructure in public and private sectors.
- Helps in reducing customer frustration by providing estimated waiting times.
- Assists management in staff allocation during peak hours.
- Can be scaled to various industries like banks, retail, hospitals, and airports.
- Provides an affordable AI solution using existing CCTV infrastructure.

Proposed Work:

1. Implement a system that uses a **camera feed + YOLO model** to detect people in real time.
2. Count the number of people in queue and calculate metrics:
    a. Queue length
    b. Average waiting time
    c. Estimated waiting time for next customer
3. Create a **web-based dashboard** (HTML/CSS/JS + Flask/FastAPI backend) to display:
    a. Live video feed with detections
    b. Queue statistics

1. *Objective*

To automatically detect and monitor queues in real-time using existing CCTV infrastructure, reducing human effort and providing accurate queue statistics such as:

- Number of people in the queue
- Average waiting time
- Alert if concedes queue length

This system helps improve customer experience and assist management in optimizing service counters.

2. *System Overview*

The system will consist of the following integrated components:

1. Camera Input Module

- o Captures live video feed using a webcam or CCTV camera.
- o Feeds the video frames into the AI detection module.

2. AI Detection Module (YOLOv5/v8)

- o Uses a pretrained YOLO (You Only Look Once) model to detect people in each frame.
- o Counts the number of detected persons (representing queue length).
- o Capable of real-time detection (~30 FPS) depending on system specs.

3. Data Processing and Analysis Module

- o Calculates queue metrics such as:
  - ▪ Queue length = Number of detected persons
  - ▪ Average waiting time = (Average service time × queue length)
  - ▪ Alert if concedes queue length = (if queueLength > upperbound)
- o Updates results continuously as new frames are processed.

4. **Backend Server (Flask/FastAPI)**
   - o Handles data flow between the AI model and the web dashboard.
   - o Processes detection results and sends them to the frontend in real-time via APIs or WebSocket.
5. **Frontend Web Dashboard (HTML/CSS/JavaScript)**
   - o Displays a live video feed with bounding boxes around detected persons.
   - o Shows real-time statistics such as:
     - ▪ Queue length
     - ▪ Average and estimated waiting time
     - ▪ Alert if concedes queue length
   - o Can be accessed by management staff on any device connected to the same network.

3. Key Functionalities

• Real-Time Detection: Detect and count people continuously.

• Dynamic Updates: Statistics update live as the queue changes.

• Visualization: Display of live detection and queue information.

• Scalability: Can be implemented across multiple service counters.

• Cost-Effective: Uses existing camera infrastructure without extra sensors.

Proposed Methodology:

1. **Problem Identification**: Need for real-time queue monitoring.
2. **Data Collection/Setup**: Use live webcam feed (or recorded video for testing)
3. **Model Integration**: Load YOLOv5/v8 pretrained model for person detection.
4. **Frame Processing**: Capture frames via OpenCV → detection → count people.
5. **Logical Calculations**: Estimate waiting time using average service duration formula.
6. **Backend Development**: Flask/FastAPI to process and send results.
7. **Frontend Dashboard**: HTML/CSS/JS dashboard with live video + charts.
8. **Testing**: Test in different environments (different lighting, crowds).

HW/SW Requirement:

Hardware:

- Standard Laptop/PC (i5 or higher, 8GB RAM)
- Webcam/External camera

Software:

- Python 3.10+
- Libraries:OpenCV,YOLOv5/v8, Flask/FastAPI, Matplotlib,pandas,scikit-learn,tensorflow
- Frontend: HTML, CSS, JavaScript
- OS: Windows/Linux

Flowchart/Algorithm:

References:

1.ChatGPT : https://chatgpt.com/

2.Applications: How AI Queue Management Are Redefining Industry Operations - Assertai

3. Model selection and evaluation — scikit-learn 1.7.2 documentation

4.Comprehensive Tutorials to Ultralytics YOLO - Ultralytics YOLO Docs

5. Get Started - OpenCV

Signature of Student

1. Satyajeet.D.More –
2. Onkar.K.Patil –
3. Prachi.I.Sakate –
4. Prashant.K.Sawant –
5. Sadabashya.A.Mujawar –

Signature of Guide **Head of Department**